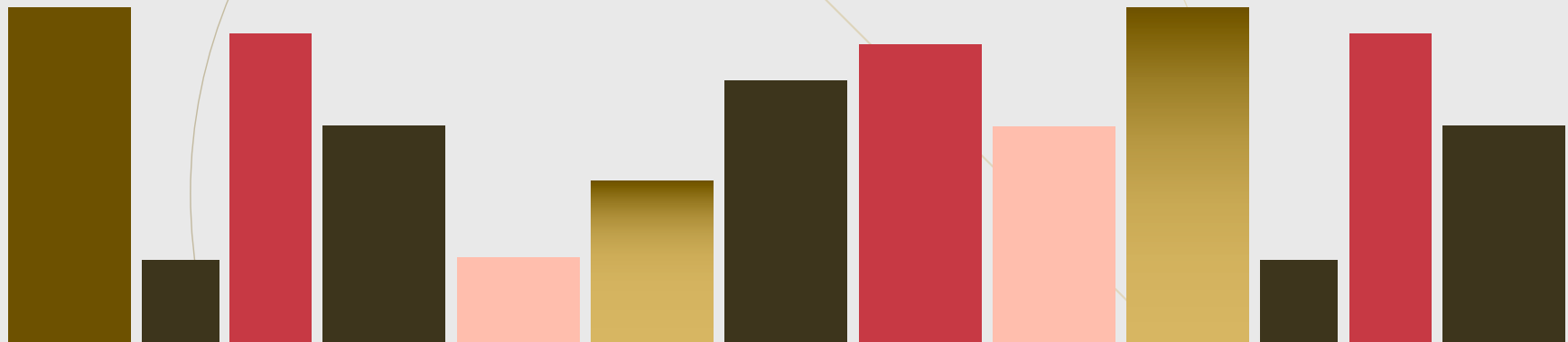


ZEN

IUT-1

10th

Exercises



25zs 04111

Slack @ luma

6. (a) B is closed in Y

$$\Leftrightarrow \exists C \in \mathcal{C},$$

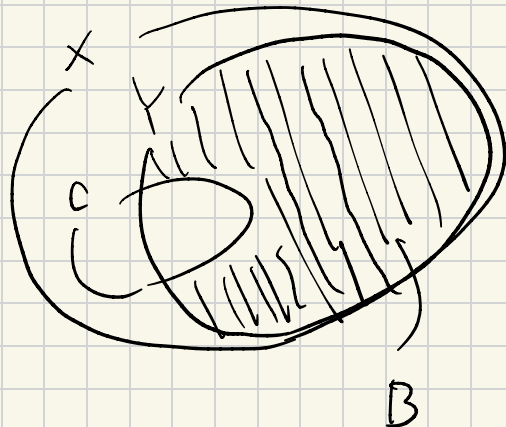
$$B = Y \setminus (C \cap Y)$$

some
induced open set

closed set
in Y

$$= (X \setminus C) \cap Y$$

closed set
in X



(b)

$$Y \cap \bigcap_{\substack{B \in X \setminus C \\ C \in \mathcal{T}}} X \setminus C = \bigcap_{\substack{B \in X \setminus C \\ C \in \mathcal{T}}} Y \cap (X \setminus C)$$

$$\begin{array}{l} B \in X \setminus C \\ C \in \mathcal{T} \end{array}$$

closed set

$$\begin{array}{l} B \in X \setminus C \\ C \in \mathcal{T} \end{array}$$

=

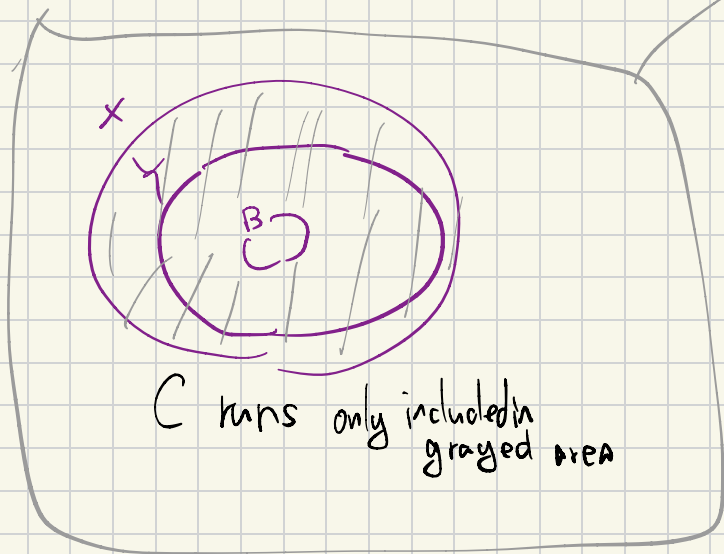
$$\bigcap_{\substack{B \in Y \setminus (Y \cap C) \\ C \in \mathcal{T}}} Y \setminus (Y \cap C)$$

$$\begin{array}{l} B \in Y \setminus (Y \cap C) \\ C \in \mathcal{T} \end{array}$$

induced open set

closed subset in Y

$$\text{cl}_Y(B)$$



7.

Statement refined

$\exists \mathcal{U}$, $(X, \mathcal{U})$ is topological space

and $S \subseteq \mathcal{U}$ and $\forall A \in \mathcal{U}$, $\exists S' \subseteq S$, $A = \bigcup S'$

$\Leftrightarrow$

$$X = \bigcup S$$

and $\forall V, W \in S \Rightarrow \exists S' \subseteq S$, $V \cap W = \bigcup S'$

(*)

□

Prf $(\Rightarrow)$ Trivial

$(\Leftarrow)$ Define $\tau := \{U S' ; S' \subseteq S\}$

(X, τ) is topological space

$\emptyset, X) \quad \emptyset \in \tau \text{ for } S' = \emptyset, \quad X \in \tau \text{ by } X = U S.$

(union) Consider $A \in \mathcal{T}$,

For any $V \in A$, there's some $S' \subseteq S$ s.t. $V = \bigcup S'$
(by def. of $\mathcal{T}$)

Let's write this as $S'(V)$

Here, let $B = \bigcup_{V \in A} S'(V)$, note that $B \subseteq S$.

$$\bigcup A = \bigcup_{V \in A} \bigcup S'(V) = \bigcup B \in \mathcal{T}$$

by def. of $\mathcal{T}$

↑ runs through all $S'(V)$

intersection) If $A, B \in \mathcal{T}$, let $A = \bigcup S_A$, $B = \bigcup S_B$,

$$A \cap B = (\bigcup S_A) \cap (\bigcup S_B) = \bigcup_{W \in S_B} (\bigcup S_A) \cap W$$

$$= \bigcup_{\substack{V \in S_A \\ W \in S_B}} V \cap W$$

Here, define $S'(v, w)$ function by (*)

$$= \bigcup_{\substack{V \in S_A \\ W \in S_B}} \underbrace{\bigcup S'(v, w)}_{\in \mathcal{T} \text{ by def. of } \mathcal{T}}$$

by (union)

$$\in \mathcal{T}$$

S is base

$S \subseteq \mathcal{T}$ by $S' = \{V\}$ for every $V \in S$.

For any $A \in \mathcal{T}$, obviously its definition of
 $\Rightarrow S' \subseteq S, A = \cup S'$



8. Let $(X, \mathcal{O}_X), (Y, \mathcal{O}_Y)$ be some topological space,
 f be some map $X \rightarrow Y$.

Statement refined

$$\forall B \in \mathcal{O}_Y, f^{-1}[B] \in \mathcal{O}_X$$

$$\Leftrightarrow \forall B \in \mathcal{O}_Y, \exists A \in \mathcal{O}_X, f^{-1}[Y \setminus B] = X \setminus A$$

□

Prf

$(\Rightarrow)$ For any $B \in \mathcal{O}_Y$, $f^{-1}[B] \in \mathcal{O}_X$ is given.

$$f^{-1}[Y \setminus B]$$

$$= f^{-1}[Y] \setminus f^{-1}[B]$$

$$= X \setminus \underbrace{f^{-1}[B]}$$

We can take this as A .

($\Leftarrow$) For any $B \in \mathcal{O}_Y$, given $\exists A \in \mathcal{O}_X$, $f^{-1}[Y \setminus B] = X \setminus A$

$$f^{-1}[Y \setminus B] = f^{-1}[Y] \setminus f^{-1}[B]$$

$$= X \setminus f^{-1}[B] = X \setminus A$$

Take $X \setminus$ both sides,

$$X \setminus (X \setminus f^{-1}[B]) = X \setminus (X \setminus A)$$

Then

$$f^{-1}[B] = A \in \mathcal{O}_X$$

Main Proof

For any finite set A in Hausdorff space X .

Define $C := \{ B ; B \text{ is open set}, B \ni x, B \cap A = \emptyset, \text{ for all } x \in X \setminus A \}$ we can take such by lemma.

$X \setminus A = \bigcup C$ and C is open sets family,

so $\bigcup C$ is open set. Therefore, A is closed. 

10. Lemma
Prf

Wazps, it's proven
in PDF as 16.3.3
(p9)

Compact set in Hausdorff space X is closed $\square$

Consider compact set A . For any $x \in X \setminus A$,

For every $x \in X \setminus A$, by Hausdorff,

let $V_a \cap W_a = \emptyset$, $V_a \ni x$, $W_a \ni a$.

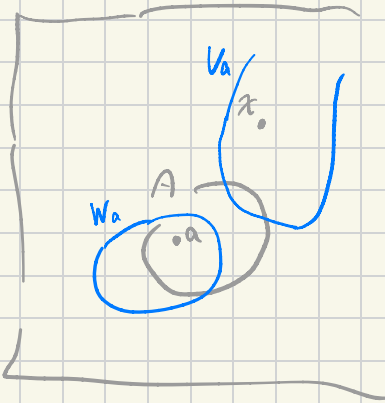
$\bigcup_{a \in A} V_a \supseteq A$ and A is compact,

so union of $V_{a_1}, V_{a_2}, \dots, V_{a_n}$ covers A .

Let $D_x := \bigcup_{i \in \mathbb{N}} V_{a_i}$ is open, and $E_x := \bigcap_{i \in \mathbb{N}} W_{a_i}$ is open.

$D_x \not\ni x$ and $D_x \supseteq A$. $E_x \ni x$ and $E_x \cap A = \emptyset$

Thus, $\bigcup_{x \in X \setminus A} D_x = X \setminus A$ and it's open. Therefore, A is closed. $\blacksquare$



Main proof

Let $f: X \rightarrow Y$, $(X, \mathcal{O}_X): \text{Compact}$, $(Y, \mathcal{O}_Y): \text{Hausdorff}$,
 f is continuous bijection.

For any closed set C of X , which is compact,
by 16.3.4 (p.9)

$f(C)$ is also compact, so closed.
by 16.5.3 (p.12) by 16.3.3 (p.9)

Then f^{-1} is continuous, and homeomorphic.



11. Lemma I

Projection of continuous is continuous.

Formally, if $f: X \times Y \rightarrow Z$ is continuous,

for any $x \in X$, $f_x: Y \rightarrow Z$, $y \mapsto f(x, y)$

is also continuous. $\square$

Prf.

For any $C \in \mathcal{O}_Z$, by f continuity,

$$\exists S_A \subseteq \mathcal{O}_X, S_B \subseteq \mathcal{O}_Y, f^{-1}(C) = \bigcup \{(A, B); A \in S_A, B \in S_B\}$$

$$f_x^{-1}(C) = \bigcup \{ B ; (A, B) \in f^{-1}(C) \text{ and } A \ni x \}$$

and this is open (union of open). 

Lemma II

In topological group, for any $A \in \mathcal{O}_G$, $g \in G \Rightarrow gA \in \mathcal{O}_G$. $\square$

Prf

Consider $f: G \rightarrow G$, $x \mapsto g^{-1}x$.

$$f^{-1}[A] = gA$$

$\swarrow$

$$gx \mapsto x$$

And f is projection of multiplication,
which is continuous, so f is also continuous, gA is open.
by lemma I $\blacksquare$

Main proof

For any $g \in G$, $\{gU; U \in \mathcal{S}\}$ is base of point g

For any $A \in \mathcal{O}_a$ and $A \ni g$, $g^{-1}A$ is including e and open too (by lemma I).

$\exists B \in \mathcal{S}$ st. $e \in B \subseteq g^{-1}A$, here $g \in gB \subseteq A$,

$gB \in \{gU; U \in \mathcal{S}\}$ and gB is open (by lemma I)

For any open set $A \in \mathcal{O}_G$, for each point $x \in A$,

we can take $B_{\alpha(x)} \in \mathcal{S}$ st. $x \in xB_{\alpha(x)} \subseteq A$.

$$xB_{\alpha(x)} \in \{gU; g \in G, U \in \mathcal{S}\} \quad \text{and} \quad \underbrace{\bigcup_{x \in A} xB_{\alpha(x)}}_{\text{Union of base}} = A$$



12. When $\prod X_i \rightarrow X_k$ is continuous, (let n be no. of spaces)
for any $A \in \mathcal{O}_{X_k}$, preimage $\{(\dots, a, \dots) ; a \in A\} = X_1 \times \dots \times A \times \dots \times X_n$
also must be open.

By intersecting them n times, for any $\{A_i \in \mathcal{O}_{X_i}\}$,

$A_1 \times A_2 \times \dots \times A_n$ is generatable by above base.

def. of base of product
(box topology)

Memo.

Unsure for infinite cases.

→ in infinite product,

Product topology $\neq$ Box topology.